

Rescue from granzyme B-induced apoptosis by Wee1 kinase.

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Granzymes are a family of granule-associated serine esterases that mediate apoptosis by cytotoxic T lymphocytes and natural killer cells. We have previously shown that cdc2, the mitosis-regulating cyclin-dependent kinase, is required for granzyme B-induced apoptosis in target cells. In addition, granzyme B induces premature activation and tyrosine dephosphorylation of cdc2 during apoptosis. Throughout most of the cell cycle and until the cell is prepared to enter mitosis, cdc2 kinase activity is negatively regulated by phosphorylation of a residue within its adenosine triphosphate-binding domain by Wee1, a nuclear kinase that maintains mitotic timing in eukaryotic cells. We have transiently expressed c-myc epitope-tagged Wee1 cDNA in BHK cells. Cells that expressed Wee1 in the nucleus became resistant to apoptosis induced by granzyme B and perforin. Wee1-transfected cells also exhibited markedly increased cdc2 tyrosine phosphorylation. Thus, Wee1 can rescue cells from granzyme-induced apoptosis by preventing cdc2 dephosphorylation.

Modulation of perforin and granzyme messenger RNA expression in human natural killer cells.

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Perforin and granzymes are proteins thought to play a relevant role in cell-mediated cytotoxicity. These molecules are constitutively expressed in NK cells and their level of expression in cytotoxic T lymphocytes is regulated by several cytokines. We analyzed the mechanisms by which cytokines and cellular ligands known to modulate NK cell-mediated cytotoxicity affect the expression of the mRNA encoding granzyme A and B and perforin in NK cells. Our data indicate that IL-2 and IL-12 induce increased accumulation of both perforin and, to a higher degree, granzyme B mRNA. In contrast, binding of target cells or immune complexes up-regulates expression of granzyme B mRNA without altering that of perforin. Results of in situ hybridization experiments confirm that mRNA for both molecules are expressed at low levels in most NK cells, and that both are induced to accumulate by the two cytokines in the majority of the cells. The mechanisms by which IL-2 and IL-12 regulate expression of the two molecules are, in part, distinct: both cytokines increase the transcriptional rate of the encoding genes, whereas only IL-2 acts also at a post-transcriptional level to increase the stability of their mRNA.

Granzyme B-H22(scFv), a human immunotoxin targeting CD64 in acute myeloid leukemia of monocytic subtypes.

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Acute myeloid leukemia (AML) cells of subtypes M4 and M5 show enhanced expression of CD64 (FcγRI), the high-affinity receptor for IgG, which is normally expressed at high levels only on activated cells of the myeloid lineage. CD64 is therefore a prime target for the specific delivery of cytotoxic agents. A promising toxin candidate is granzyme B, a human serine protease originating from cytotoxic granules of CD8+ T lymphocytes and natural killer cells. After evaluating the sensitivity of the AML-related cell line U937 toward cytosolic granzyme B, we genetically fused granzyme B to H22, a humanized single-chain antibody fragment (scFv) specific for CD64, to obtain Gb-H22(scFv), a fusion

protein lacking the immunogenic properties of nonhuman immunofusions. Gb-H22(scFv) was successfully expressed in human 293T cells, secreted, and purified from cell culture supernatants. The purified protein bound specifically to CD64+ U937 cells. Despite linkage to the binding domain, the proteolytic activity of functional Gb-H22(scFv) was identical to that of free granzyme B. Target cell-specific cytotoxicity was observed with a half-maximal inhibitory concentration (IC50) between 1.7 and 17 nmol/L. In addition, the induction of apoptosis in U937 cells was confirmed by Annexin A5 staining and the detection of activated caspase-3 in the cytosol. Finally, apoptosis was observed in primary CD64+ AML cells, whereas CD64(-) AML cells were unaffected. This is the first report of a completely human granzyme B-based immunotoxin directed against CD64, with activity against an AML-related cell line and primary AML cells.

Plaques en prairie fauchée.

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We report a clinical picture of a mowed meadow pattern of the tongue in a 29-year-old sexually active man who has sex with men. The painful lesions also called "Plaques en prairie fauchée", occur in the context of the discovery of a positive HIV serology. The diagnosis of secondary syphilis was confirmed by a positive syphilis serology (nontreponemal RPR test titer of 64) and a positive *Treponema pallidum* PCR, amplifying the *tp47* gene, on a tongue swab. The lesions completely resolve with a penicillin treatment.

Secondary Syphilis Presenting as Glossodynia, Plaques en Prairie Fauchée, and a Split Papule at the Oral Commissure: Case Report and Review.

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Syphilis has been coined "the great imitator" due to its extreme heterogeneity of presentation and mimicry of other conditions. Therefore, it is essential that physicians be familiar with the full spectrum of its manifestations. Syphilis may also lead to oral lesions that, occasionally, are unaccompanied by concomitant tegumentary findings. Such patients will pose unique diagnostic challenges. We report the case of a 45-year-old HIV-infected male patient in whom secondary syphilis presented with burning mouth and dysgeusia that progressed to glossodynia and odynophagia. Examination revealed painful, shallow erosions on the posterior aspect of the tongue, in a pattern of plaques en prairie fauchée. A painful split papule (fausse perlèche or false angular cheilitis) was also present in the left commissure. There were no cutaneous lesions. The oral lesions were considered highly suggestive of secondary syphilis. A novel VDRL assay (which was previously negative) yielded a titer of 1/128. Complete clinical remission was rapidly achieved after initiation of penicillin therapy. A comprehensive review of the literature on oral manifestations of syphilis is offered.

Gender disparities and predictors of in-hospital mortality with Takotsubo cardiomyopathy.

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Takotsubo cardiomyopathy (TCM) is classically associated with significant gender disparities, such that it

is more prevalent in females, but the clinical outcomes are worse for male patients. The goal of this study was to assess contemporary gender disparities in clinical outcomes of TCM hospitalizations and to determine predictors of male in-hospital mortality. This was a retrospective analysis involving adult hospitalizations for TCM in the U.S between 2016 and 2020. Multivariable Logistic regression was used to estimate Odds Ratio (OR) for in-hospital mortality between the two genders. Univariable Cox regression was performed to identify predictors associated with in-hospital mortality for male hospitalizations. All factors from the univariable analysis with $p < 0.20$ were included in a multivariable Cox regression model. A total of 199,920 patients with TCM were identified. Female patients with TCM had 50% lower risk of in-hospital mortality compared to male patients (Adjusted OR 0.50, 95% CI 0.46-0.55, $p < 0.001$). Older age, higher Charlson comorbidity index, history of intracranial hemorrhage, cardiac arrest, need for vasopressor agents, mechanical intubation, and cardiogenic shock without the use of temporary mechanical circulatory support (MCS) were associated with higher in-hospital male mortality. Although TCM is more prevalent among females, gender disparities exist in the clinical outcomes of TCM patients. Cardiac arrest and cardiogenic shock without the use of temporary MCS were found to be the most significant predictors of male in-hospital mortality. Cardiogenic shock with use of temporary MCS did not lead to higher male in-hospital mortality.

Sex Differences in the Clinical Outcomes of Patients With Takotsubo Stress Cardiomyopathy: A Meta-Analysis of Observational Studies.

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The incidence of takotsubo stress cardiomyopathy (TSCM) in males is low compared with females. Gender-based differences in clinical outcomes of TSCM are not well characterized. The aim of this meta-analysis was to analyze whether gender-based differences are observed in TSCM clinical outcomes. A comprehensive literature search of PubMed, Embase, Cochrane Library database, and Web of Science was performed from inception to June 20, 2022, for studies comparing the clinical outcomes between male and female patients with TSCM. The primary outcome of interest was in-hospital all-cause mortality and cardiogenic shock. The secondary outcomes were cardiovascular mortality, receipt of mechanical ventilation, intra-aortic balloon pump, occurrence of ventricular arrhythmia, and left ventricular thrombus. A random-effects model was used to calculate the risk ratios (RR) and confidence intervals (CI). Heterogeneity was assessed using the Higgins I² index. Twelve observational studies involving 51,213 patients (4,869 males and 46,344 females) were included in the meta-analysis. Male gender was associated with statistically significant higher in-hospital all-cause mortality compared with females in patients with TSCM (RR 2.17, 95% CI 1.77 to 2.67, $p < 0.001$). The rate of cardiogenic shock was significantly higher in males with TSCM compared with females (RR 1.66, 95% CI 1.29 to 2.12, $p < 0.001$). Our meta-analysis showed a difference in the clinical outcomes of TSCM between men and women. Male gender was associated with a two-fold greater in-hospital all-cause mortality risk compared with female gender. The higher mortality risk associated with male gender deserves further study, particularly whether it represents later recognition of the condition and disparities in treatments.

Takotsubo Syndrome and Gender Differences: Exploring Pathophysiological Mechanisms and Clinical Differences for a Personalized Approach in Patient Management.

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Takotsubo syndrome (TTS), also known as the broken-heart syndrome, is a reversible condition typically observed in female patients presenting for acute coronary syndromes (ACS). Despite its increasing incidence, TTS often remains undiagnosed due to its overlap with ACS. The pathophysiology of TTS is complex and involves factors such as coronary vasospasm, microcirculatory dysfunction, increased catecholamine levels, and overactivity of the sympathetic nervous system. Diagnosing TTS requires a comprehensive approach, starting with clinical suspicion and progressing to both non-invasive and invasive multimodal tests guided by a specific diagnostic algorithm. Management of TTS should be personalized, considering potential complications, the presence or absence of coronary artery disease (CAD), diagnostic test results, and the patient's clinical course. The current data primarily derive from case series, retrospective analyses, prospective registries, and expert opinions. In recent years, there has been growing recognition of gender differences in the pathophysiology, presentation, and outcomes of TTS. This review provides an updated overview of gender disparities, highlighting the importance of tailored diagnostic and management strategies.

Gender disparities in outcomes of cardiogenic shock secondary to Takotsubo cardiomyopathy.

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Takotsubo cardiomyopathy (TTC) has a preponderance for females, particularly postmenopausal. However, recent data from multicenter registries identified a worse prognosis in male patients, particularly with cardiogenic shock. We aim to investigate gender disparities in outcomes of TTC-associated cardiogenic shock (TTC-CS). The National Readmission Database (2016-2020) was utilized to identify TTC-CS hospitalizations. Cohorts were stratified by gender. A Propensity Score Matching (PSM) model, which utilized complete Mahalanobis Distance Matching within the Propensity Score Caliper following multivariate regression, successfully matched males and females. Pearson's χ^2 test was applied to the propensity-matched cohorts to compare outcomes. Among 12,803 TTC-CS hospitalizations, the majority (74.1 %) were females (N: 9490), and 25.9 % were males (N: 3313). On propensity-matched cohorts (2609), males were found to have higher in-hospital mortality (31 % vs. 26 %, $p < 0.001$), higher incidence of sudden cardiac arrest (14 % vs. 10.8 %, $p < 0.001$), endotracheal intubation (52.1 % vs. 48.8 %, $p: 0.001$), acute liver injury (18 % vs. 15.9 %, $p: 0.004$), acute stroke (7.2 % vs. 5.8 %, $p: 0.004$), cardiac arrhythmias (55.1 % vs. 49.3 %, $p < 0.001$) and acute kidney injury (63.1 % vs. 49 %, $p < 0.001$); while female patients were found to have higher utilization of mechanical circulatory support (MCS) modalities (16.1 % vs 13.2 %, $p < 0.001$). Males had a higher adjusted cost of hospitalization (\$54,537 vs. \$42,805, $p < 0.001$) with a higher median length of hospital stay (10 vs. 9 days, $p < 0.001$). The two groups had no significant difference in 30, 90, and 180-day readmission rates ($p > 0.05$). From 2016 to 2020; mortality has not changed significantly for TTC-CS, while the use of percutaneous coronary angiogram (PCA) and MCS has down-trended (p -trend < 0.05). For TTC-CS hospitalization, males have higher in-hospital mortality and complication rates, along with higher LOS and cost of hospitalization. Despite advances in the management of CS, there was no significant difference in mortality from 2016 to 2020.

Takotsubo Syndrome and Sudden Cardiac Death.

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Takotsubo syndrome (TTS), triggered by intense emotional or physical stress, occurring most commonly in post-menopausal women, presents as an ST-elevation myocardial infarction (MI). Cardiovascular complications occur in almost half the patients with TTS, and the inpatient mortality is comparable to MI (4-5%) owing to cardiogenic shock, myocardial rupture, or life-threatening arrhythmias. Thus, its prognosis is not as benign as previously thought, as it may cause mechanical complications (cardiac rupture) and potentially lethal arrhythmias and sudden cardiac death (SCD). Similar to MI, some patients may perish before reaching the hospital due to out-of-hospital cardiac arrest; this may lead to underestimation of the actual SCD risk. Furthermore, after discharge, some patients may develop late SCD and/or TTS recurrence that may result in SCD. There are risk factors for SCD in TTS patients, such as severe/persistent QT-interval prolongation inciting torsade-de-pointes, other ECG abnormalities (diffuse giant negative T-waves, widened QRS-complex), bradyarrhythmias, comorbidities, concurrent obstructive coronary artery disease or vasospasm, male gender, older age, severe left ventricular dysfunction, and use of sympathomimetic drugs. All these issues are herein reviewed, case reports/series and data from large cohort studies and meta-analyses are analyzed, risk factors are tabulated, and proarrhythmic effects and management strategies are discussed and pictorially illustrated.